Comprehensive Model Results Summary

Hedonic Price Index Analysis

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14 Models: 7 Types × 2 Time Dimensions (Quarterly/Annual)

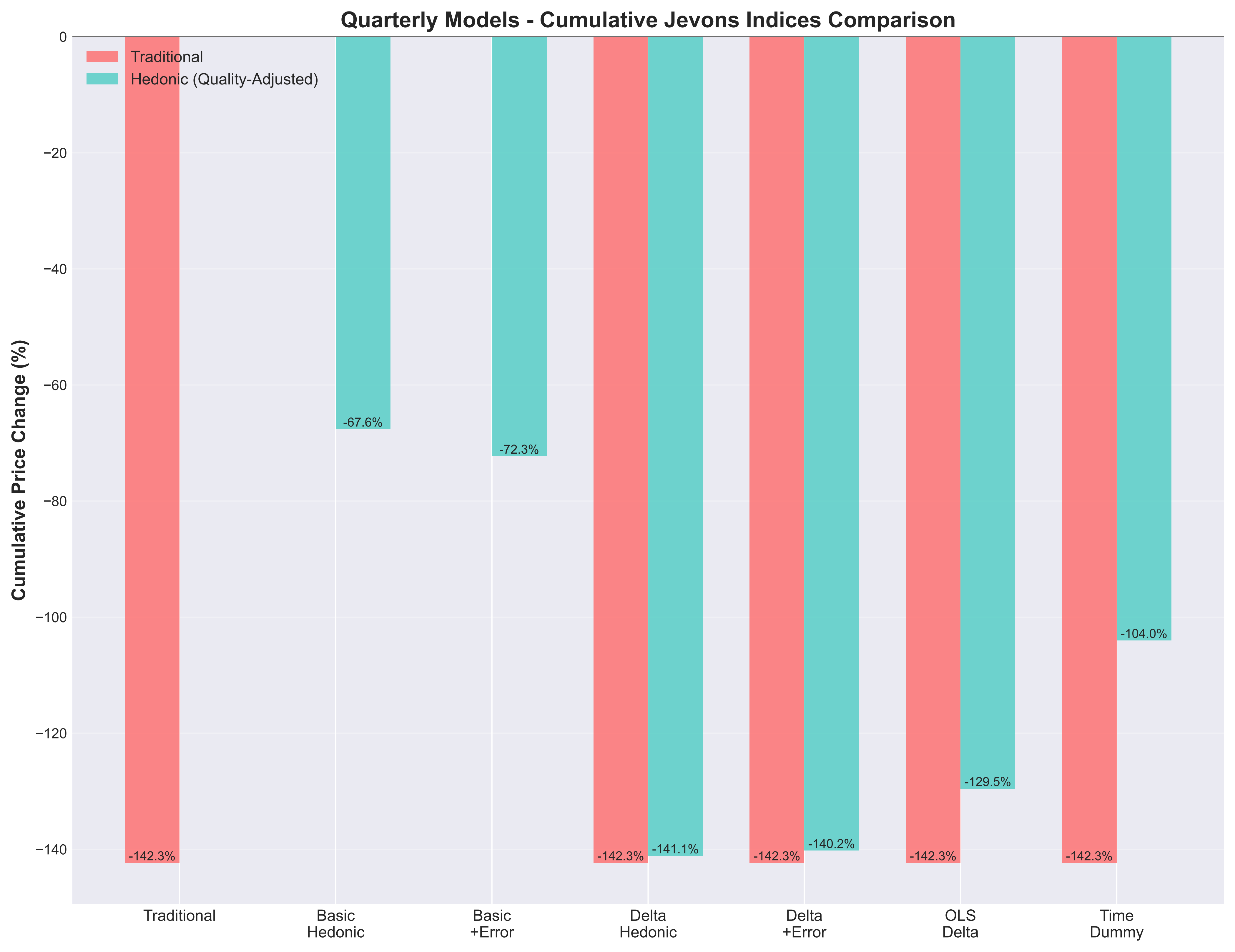
Quarterly Models - Cumulative Jevons Indices

Model	Final Chained Jevons Index	Cumulative Deflation	R^2 (price-change)
Traditional Jevons	-1.4231	-75.90%	N/A
Levels Hedonic	-0.6759	-49.13%	0.1279
Levels Hedonic with Lagged	-0.7226	-51.45%	0.0936
Price-Change Hedonic	-1.4110	-75.61%	0.2559
Price-Change Hedonic with	-1.4020	-75.39%	0.2772
Price-Change Hedonic with	-1.2952	-72.62%	0.4364
Levels Hedonic with Time-D	-1.0399	-64.65%	0.0000

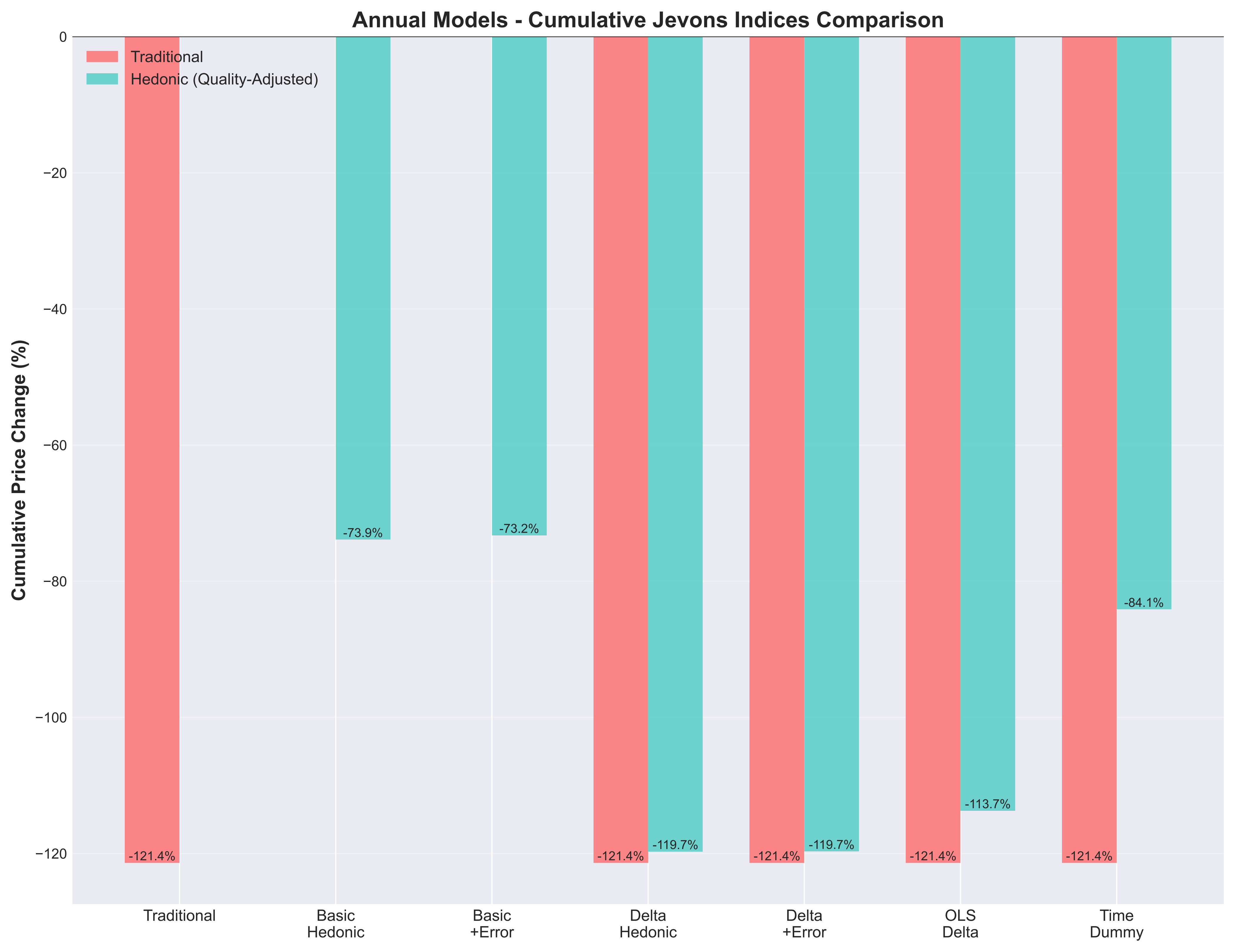
Annual Models - Cumulative Jevons Indices

All models cover 2020 to 2025 (5 periods)

Model	Final Chained Jevons Index	Cumulative Deflation	R^2 (price-change)
Traditional Jevons	-1.2135	-70.29%	N/A
Levels Hedonic	-0.7387	-52.22%	0.1915
Levels Hedonic with Lagged	-0.7325	-51.93%	0.1012
Price-Change Hedonic	-1.1970	-69.79%	0.2498
Price-Change Hedonic with	-1.1968	-69.79%	0.2623
Price-Change Hedonic with	-1.1371	-67.92%	0.4226
Levels Hedonic with Time-D	-0.8412	-56.88%	0.0000



Quarterly Models - Cumulative Jevons Indices Comparison
Traditional
Hedonic (Quality-Adjusted)
Cumulative Price Change (%)
0
−20
−40
−60
−80
−100
−120
−140
-142.3%
-67.6%
-72.3%
-142.3%
-141.1%
-142.3%
-140.2%
-142.3%
-129.5%
-142.3%
-104.0%
Traditional
Basic Hedonic
Basic +Error
Delta Hedonic
Delta +Error
OLS Delta
Time Dummy


Annual Models - Cumulative Jevons Indices Comparison
Traditional
Hedonic (Quality-Adjusted)
Cumulative Price Change (%)
0
-20
-40
-60
-80
-100
-120
-121.4%
-73.9%
-73.2%
-121.4%
-119.7%
-121.4%
-119.7%
-121.4%
-113.7%
-121.4%
-84.1%
Traditional
Basic Hedonic
Basic +Error
Delta Hedonic
Delta +Error
OLS Delta
Time Dummy

Model Descriptions

- **Traditional Jevons Index:**
  Direct calculation using actual prices. No quality adjustment. Serves as baseline.

- **Basic Hedonic:**
  Standard hedonic regression. Independent Lasso model for each period. Controls for product features.

- **Basic + Error Feature:**
  Hedonic regression with previous period prediction error as additional feature. Captures time dependencies.

- **Delta Model (Lasso):**
  Directly models log price differences between consecutive periods using Lasso. Uses OOF predictions to avoid mean-matching artifacts. Automatic feature selection.

- **Delta + Error Feature:**
  Combines Delta model with error feature from previous period pair. Captures both price changes and time dependencies.

- **OLS Delta Model:**
  Directly models log price differences using OLS (no regularization). All features included. Simpler interpretation.

- **Time Dummy Model:**
  Pools consecutive periods data together. Adds time dummy variable. Single model predicts both periods. More efficient parameter sharing.